

AN *APHELASPIS* ZONE (UPPER CAMBRIAN, PAIBIAN) TRILOBITE FAUNULE IN THE CENTRAL CONASAUGA RIVER VALLEY, NORTH GEORGIA, USA

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ABSTRACT

Middle and Upper Cambrian strata (Cambrian Series 3 and Furongian) in the southernmost Appalachians (Tennessee to Alabama) comprise the Conasauga Formation or Group. Heretofore, the youngest reported Conasauga beds in the Valley and Ridge Province of Georgia were of the late Middle Cambrian (Series 3: Drumian) *Bolaspidella* Zone, located on the western state boundary in the valley of the Coosa River. Two new localities sited eastward in the Conasauga River Valley, yield a diagnostic suite of trilobites from the Upper Cambrian *Aphelaspis* Zone. Very abundant, polymeroid trilobites at the primary locality are referable to *Aphelaspis brachyphasis*, which is a species known previously in western North America. A second locality has produced a few identifiable specimens of the aphelaspine *Eugonocare* (*Olenaspella*) *separatum*. Specimens at these two localities are generally complete individuals compressed in tan mudstones. The primary locality features abundant body cluster accumulations, implying mass mortality by bioimmuration. The trilobite assemblage also includes the agnostoids *Glyptagnostus reticulatus*, *Agnostus inexpectans*, and *Aspidagnostus rugosus*, all correlated to the global Paibian agnostoid *Glyptagnostus reticulatus* Zone. These localities contain the southeastern-most Late Cambrian faunule in the Appalachians. The trilobite species and carbonate-free, mudstone lithology, lacking evidence of infaunal bioturbation and burrowing, suggest accumulation eastward of a paleotopographic

shelf-to-basin break, which is interpreted to be east of the Alabama Promontory and in the Tennessee Embayment. The preservation of abundant aphelaspine specimens by bioimmuration events may have been the result of mudflows down the shelf-to-basin slope.

INTRODUCTION

Trilobites and associated biota from Middle Cambrian beds of the Conasauga Formation in northwestern Georgia have been described by Walcott, 1916a, 1916b; Butts, 1926; Resser, 1938; Palmer, 1962; Schwimmer, 1989; Schwimmer and Montante, 2007. These fossils and deposits come from exposures within the valley of the Coosa River, in Floyd County, Georgia, and adjoining Cherokee County, Alabama. Trilobite biozone associations of these Middle Cambrian biotas are of the *Glyphaspis* to *Crepicephalus* Zones of the traditional Laurentian Middle Cambrian Series.

Cambrian trilobites have not been described farther east- or southward from the above in the southern Appalachian outcrop, although the Conasauga Formation is exposed in a separate fault-bounded slice in the Conasauga River Valley (Figure 1). We report here new Upper Cambrian localities in this outcrop, containing the southeastern-most Cambrian trilobites in North America. The new trilobite assemblages include polymeroids of the Laurentian Upper Cambrian *Aphelaspis* Zone, including *Aphelaspis brachyphasis* and *Eugonocare* (*Olenaspella*) *separatum*; and coeval agnostoids of the global *Glyptagnostus reticulatus* Zone, including *Glyptagnostus reticulatus*, *Agnostus in-*